

15/04/2024



Phase-I  
CODE-A

# Aakash

Medical | IIT-JEE | Foundations

Corporate Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

## FINAL TEST SERIES for NEET-2024

MM : 720

Test -8

Time : 3 Hrs. 20 Mins.

### Topics covered :

**Physics** : Dual Nature of Radiation and Matter, Atoms, Nuclei, Semiconductor Devices, Experimental Skills

**Chemistry** : Amines, Biomolecules, Principles Related to Practical Chemistry

**Botany** : Ecosystem, Biodiversity and Conservation

**Zoology** : Biotechnology: Principles and Processes, Biotechnology and Its Applications

### Instructions :

- There are two sections in each subject, i.e. Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions from Section-B out of 15.
- Each question carries 4 marks. For every wrong response 1 mark shall be deducted from the total score. Unanswered / unattempted questions will be given no marks.
- Use blue/black ballpoint pen only to darken the appropriate circle.
- Mark should be dark and completely fill the circle.
- Dark only one circle for each entry.
- Dark the circle in the space provided only.
- Rough work must not be done on the Answer sheet and do not use white-fluid or any other rubbing material on the Answer sheet.

## PHYSICS

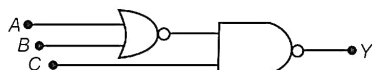
Choose the correct answer :

### SECTION - A

1. Minimum number of NOR gates are used to form an OR gate?

- (1) 1 (2) 2  
(3) 3 (4) 4

2. The output Y, when all three inputs are first high and then low, will respectively be



- (1) 1, 0  
(2) 1, 1  
(3) 0, 0  
(4) 0, 1

3. Electrons are emitted with zero velocity from a metal surface when it is exposed to radiation of wavelength 6200 Å. Work function ( $\phi_0$ ) of the metal is

- (1) 1 eV (2) 2 eV  
(3) 2.5 eV (4) Zero

4. If the wavelength of electromagnetic radiation is doubled, then the energy of photon corresponding to above radiation will be

- (1) Doubled  
(2) Halved  
(3) Remains same  
(4) Four times the initial value

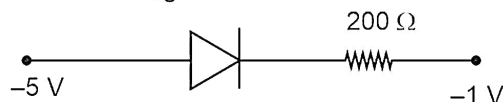
5. Match the entries in the column I with corresponding correct entries in column II. (Where symbols have their usual meaning)

|     | Column I                        |       | Column II                  |
|-----|---------------------------------|-------|----------------------------|
| (A) | de-Broglie wavelength           | (i)   | $K_{\max} = h\nu - \phi_0$ |
| (B) | Threshold frequency             | (ii)  | $\lambda = \frac{h}{p}$    |
| (C) | Einstein photoelectric equation | (iii) | $v = \frac{\phi_0}{h}$     |

- (1) (A)→(iii), (B)→(i), (C)→(ii)  
 (2) (A)→(ii), (B)→(iii), (C)→(i)  
 (3) (A)→(iii), (B)→(ii), (C)→(i)  
 (4) (A)→(ii), (B)→(i), (C)→(iii)
6. Electron volt (eV) is the unit of  
 (1) Energy (2) Potential  
 (3) Charge (4) Current
7. Which of the following force is responsible for  $\alpha$ -particle scattering?  
 (1) Gravitational (2) Nuclear  
 (3) Coulomb (4) Magnetic
8. The ground state energy of an electron in a hydrogen atom is  $-13.6$  eV. What is the potential energy of the electron in this state?  
 (1)  $-27.2$  eV (2)  $-13.6$  eV  
 (3)  $+13.6$  eV (4)  $+27.2$  eV
9. The energy equivalent to 5 milligrams of matter is  
 (1)  $45 \times 10^{13}$  J (2)  $45 \times 10^{10}$  J  
 (3)  $5 \times 10^{10}$  J (4)  $4.5 \times 10^{10}$  J
10. Ratio of radii of  $\text{Li}^7$  nucleus to  $\text{Fe}^{56}$  nucleus will be  
 (1) 1 : 8 (2) 7 : 8  
 (3)  $\sqrt{\frac{7}{56}}$  (4) 1 : 2

11. A N-type semiconductor material is electrically  
 (1) Positive  
 (2) Negative  
 (3) Neutral  
 (4) Depends on the concentration of the impurities

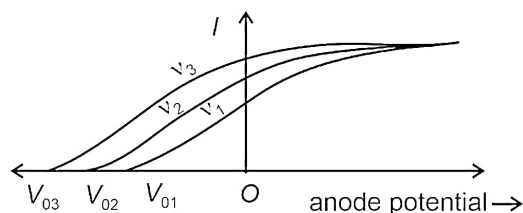
12. Current through the diode as shown below will be



- (1) 20 mA (2) 20 A  
 (3) 30 mA (4) Zero
13. We can increase the sensitivity of a moving coil galvanometer by  
 (1) Increasing the area of the coil  
 (2) Increasing the magnetic field  
 (3) Decreasing the torsional constant  
 (4) All of these
14. If the de-Broglie wavelengths for a proton and a deuteron are equal, then the ratio of their velocities will be  
 (1) 2 : 1 (2) 4 : 1  
 (3) 1 : 3 (4) 1 : 4
15. Which of the following is correct relation for vernier constant (V.C.)? (Here symbols have their usual meaning)  
 (1) V.C. = 1 VSD + 1 MSD  
 (2) V.C. = 1 MSD - 1 VSD  
 (3) V.C. =  $\frac{1\text{MSD}}{1\text{VSD}}$   
 (4) V.C. = 1 MSD  $\times$  1 VSD
16. In resonance tube first resonating length is 25 cm and second resonating length is 80 cm, then the value of end correction will be  
 (1) 5 cm (2) 2.5 cm  
 (3) 15 cm (4) 27.5 cm

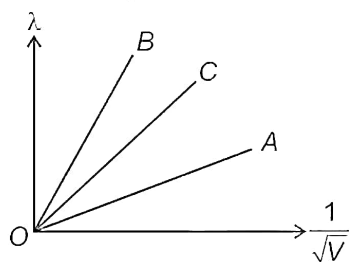
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17. The effect of frequency of incident radiation on stopping potential at a particular intensity of light radiation is shown



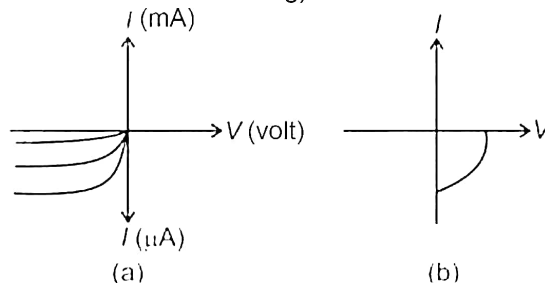
Which of the following relation may be correct based upon above information?

- (1)  $\nu_1 > \nu_2 < \nu_3$  (2)  $\nu_2 = \nu_3 < \nu_1$   
 (3)  $\nu_3 > \nu_2 > \nu_1$  (4)  $\nu_1 = \nu_2 = \nu_3$
18. The graph shown below shows variation of de Broglie wavelength ( $\lambda$ ) versus  $\frac{1}{\sqrt{V}}$  (where  $V$  is accelerating potential) for three charged particles of equal charges but different masses. Tick the correct option.



- (1)  $m_A > m_B > m_C$  (2)  $m_B > m_C > m_A$   
 (3)  $m_A > m_C > m_B$  (4)  $m_B > m_A > m_C$
19. Consider a metal surface which is able to eject photoelectrons when visible light is incident over it. Which of the following has maximum stopping potential when metal is illuminated by visible light?
- (1) Red (2) Yellow  
 (3) Blue (4) Violet

20. A photoelectric cell converts  
 (1) Light energy into heat energy  
 (2) Heat energy into light energy  
 (3) Light energy into electrical energy  
 (4) Electrical energy into light energy
21. Which of the following statements is incorrect about photoelectric effect?  
 (1) It is based on particle nature of light  
 (2) Stopping potential depends on the intensity of incident light having fixed frequency  
 (3) Saturation current depends on the intensity of incident light having fixed frequency  
 (4) Frequency should be greater than threshold frequency for emission of electrons
22. An electron is accelerated through a potential difference of 64 V. de Broglie wavelength associated with it will be nearly  
 (1) 4.65 Å (2) 1.53 Å  
 (3) 2.10 Å (4) 3.56 Å
23. A p-n junction photodiode is fabricated from a semiconductor with band gap of 2.2 eV. It can detect a light of wavelength  
 (1) 6000 nm (2) 5000 Å  
 (3) 700 nm (4) 6000 Å
24. The  $I$ - $V$  characteristics for two semiconductor devices are shown in figure (Where symbols have their usual meaning)

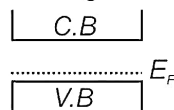


The corresponding devices are

- (1) LED, Solar cell  
 (2) LED, Photodiode  
 (3) Photodiode, Solar cell  
 (4) Photodiode, Zener diode

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25. The energy band diagram of a semiconductor material is shown in figure.



The semiconductor is of

- (1) n-type (2) p-type  
(3) Intrinsic type (4) Data is insufficient
26. The radius of second stationary orbit in Bohr's atom is  $R$ . The radius of the third stationary orbit in the same atom will be  
(1)  $3R$  (2)  $9R$   
(3)  $\frac{5R}{4}$  (4)  $\frac{9R}{4}$
27. Among the given options, the set which represents isotopes, isobars and isotones respectively, is  
(1)  $({}_1\text{H}^2, {}_1\text{H}^3), ({}_{79}\text{Au}^{197}, {}_{80}\text{Hg}^{198}) \& ({}_2\text{He}^3, {}_1\text{H}^2)$   
(2)  $({}_1\text{H}^2, {}_1\text{H}^3), ({}_2\text{He}^3, {}_1\text{H}^3) \& ({}_{79}\text{Au}^{197}, {}_{80}\text{Hg}^{198})$   
(3)  $({}_2\text{He}^3, {}_1\text{H}^1), ({}_1\text{H}^2, {}_1\text{H}^3) \& ({}_{79}\text{Au}^{197}, {}_{80}\text{Hg}^{198})$   
(4)  $({}_2\text{He}^3, {}_1\text{H}^1), ({}_{79}\text{Au}^{197}, {}_{80}\text{Hg}^{198}) \& ({}_1\text{H}^1, {}_1\text{H}^3)$
28. Inside a nucleus, if nuclear force between two protons and between a proton and a neutron are  $F_{pp}$  and  $F_{pn}$  respectively, then  
(1)  $F_{pp} < F_{pn}$  (2)  $F_{pp} > F_{pn}$   
(3)  $F_{pp} = F_{pn}$  (4)  $F_{pp} = 131 F_{pn}$
29. A lithium nucleus has three protons and four neutrons. If the mass of Li nucleus is 7.016005 amu and mass of proton is 1.007277 amu and mass of neutron is 1.008665 amu, then mass defect of lithium nucleus in amu is  
(1) 0.05010 (2) 0.4050  
(3) 0.9021 (4) 0.040486

30. Choose the incorrect statement.

- (1) The emission of electron in  $\beta^-$  decay is accompanied by the emission of antineutrino  
(2) The emission of electron in  $\beta^+$  decay is accompanied by the emission of neutrino  
(3) Free neutron unlike a free proton, is unstable  
(4) In  $\beta$  decay atomic number  $Z$  remains unchanged
31. At room temperature pure Si have equal number of electron and hole concentration of  $1.5 \times 10^{10} \text{ cm}^{-3}$ . Doping by a trivalent atom, hole concentration increases to  $4.5 \times 10^{22} \text{ m}^{-3}$ . The electron concentration of the semiconductor will be  
(1)  $5 \times 10^3 \text{ m}^{-3}$  (2)  $5 \times 10^{-2} \text{ m}^{-3}$   
(3)  $5 \times 10^9 \text{ m}^{-3}$  (4)  $2.5 \times 10^9 \text{ m}^{-3}$
32. A nucleus  ${}_n^m\text{X}$  emits two alpha particles and two beta ( $\beta^-$ ) particles. The resulting nucleus is  
(1)  ${}_{n-2}^{m-4}\text{Y}$  (2)  ${}_{n-1}^{m-8}\text{Y}$   
(3)  ${}_{n-2}^{m-8}\text{Y}$  (4)  ${}_{n+2}^{m-8}\text{Y}$
33. The binding energies of nuclei  $X$  and  $Y$  are  $E_1$  and  $E_2$  respectively. Two atoms of  $X$  fuse to give one atom of  $Y$  and an energy  $Q$  is released, then  
(1)  $Q = 2E_1 - E_2$  (2)  $Q = E_2 - 2E_1$   
(3)  $Q = E_1 + E_2$  (4)  $Q = E_1 - E_2$
34. Which of the following gate corresponds to the given truth table?
- | A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |
- (1) NOR (2) NAND  
(3) AND (4) OR

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35. The statement which is **incorrect** for nuclear force between two protons is
- These are always attractive
  - These are non-central forces
  - These are charge independent
  - These are short range forces

**SECTION - B**

36. N-type semiconductor is formed when
- P impurity is mixed in Si
  - Al impurity is mixed in Si
  - Ga impurity is mixed in Ge
  - As impurity is mixed in Ge
- (i) and (ii)
  - (ii) and (iii)
  - (ii) and (iv)
  - (i) and (iv)
37. Read the following statements and select the **correct** option.
- Assertion (A):** The kinetic energy of photoelectrons emitted from metal surface does not depend on the intensity of incident photon.
- Reason (R):** In process of photoelectric emission, all emitted electrons do not have same kinetic energy.
- Both A and R are true and R is the correct explanation of A
  - Both A and R are true but R is not the correct explanation of A
  - A is true but R is false
  - Both A and R are false
38. When the kinetic energy of an electron is decreased, the de-Broglie wavelength of the associated wave will
- Increase
  - Decrease
  - Remain same
  - May increase or decrease

39. Stopping potential for a fast-moving photo-electron is independent of
- Frequency of incident photon
  - The intensity of incident photon
  - The wavelength of an incident photon
  - Both (1) and (3)
40.  $m_p$  and  $m_n$  are masses of proton and neutron respectively. A stable element of nuclear mass  $M$  has  $Z$  protons and  $N$  neutrons then
- $M > Zm_p + Nm_n$
  - $M < Zm_p + Nm_n$
  - $M = Zm_p + Nm_n$
  - $M > 2Zm_p$
41. In a fission process, nucleus  $A$  divides into two nuclei  $B$  and  $C$ , their binding energies are  $E_a$ ,  $E_b$  and  $E_c$  respectively. Then
- $E_b + E_c < E_a$
  - $E_b + E_c = E_a$
  - $E_b + E_c > E_a$
  - $E_a = 2E_b + 3E_c$
42. The energy of an excited hydrogen atom is  $-1.51$  eV. The angular momentum of the electron in that orbit according to Bohr's model will be
- $\frac{h}{2\pi}$
  - $\frac{h}{\pi}$
  - $\frac{3h}{2\pi}$
  - $\frac{3h}{\pi}$
43. In which of the following transition of electron in hydrogen atom will the wavelength be minimum?
- $n = 5$  to  $n = 3$
  - $n = 4$  to  $n = 2$
  - $n = 4$  to  $n = 3$
  - $n = 3$  to  $n = 1$
44. The Young's modulus of a wire of radius  $R$  and length  $L$  is  $Y$ . If the radius and length are changed to  $\frac{R}{2}$  and  $2L$  respectively, then its Young's modulus will be
- $Y$
  - $4Y$
  - $8Y$
  - $2Y$
45. The circular scale of a screw gauge contains 100 divisions and its pitch is 1 mm. The least count of the screw gauge is
- 0.01 mm
  - 0.1 mm
  - 0.05 mm
  - 1 mm

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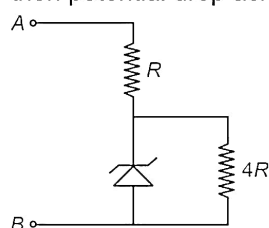
46. To determine refractive index of glass slab using a travelling microscope, minimum number of readings required are

- (1) Four (2) Six  
(3) Three (4) Two

47. The work function of a material is 5.3 eV and the stopping potential for a radiation incident on this surface is 4.2 V. The energy of a quantum of incident radiation is

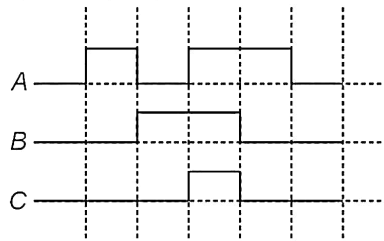
- (1) 1.1 eV (2) 9.5 eV  
(3) 5.3 eV (4) 4.2 eV

48. If the potential difference between terminal A and B is 15 V and Zener breakdown voltage is 10 V, then potential drop across R is



- (1) 5 V (2) 6 V  
(3) 7 V (4) 10 V

49. The waveform of input A and B and the output C are given below. The logic gate which gives the corresponding output is



- (1) OR gate  
(2) NOR gate  
(3) NAND gate  
(4) AND gate

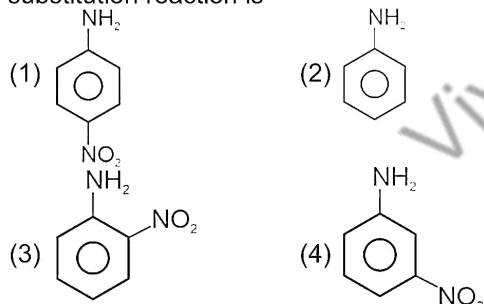
50. The de-Broglie's wavelength associated with a particle of mass  $m = 2 \text{ kg}$  and moving with a velocity  $100 \text{ m/s}$  is nearly

- (1)  $6.6 \times 10^{-34} \text{ m}$  (2)  $3.3 \times 10^{-36} \text{ m}$   
(3)  $3.3 \times 10^{-34} \text{ m}$  (4)  $5 \times 10^{-37} \text{ m}$

## CHEMISTRY

### SECTION - A

51. Most reactive species towards electrophilic substitution reaction is



52. Number of primary and secondary amines possible for the molecular formula  $\text{C}_4\text{H}_{11}\text{N}$  respectively (excluding stereoisomers) are

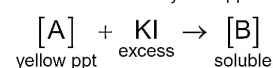
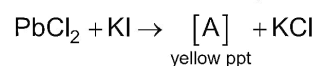
- (1) 4 and 4 (2) 5 and 4  
(3) 4 and 3 (4) 3 and 3

53. Salt +  $\text{AgNO}_3 \rightarrow$  (A) Yellow ppt

A is almost insoluble in concentrated  $\text{NH}_3$  solution. Identify the anion in the salt.

- (1)  $\text{Cl}^-$  (2)  $\text{S}^{2-}$   
(3)  $\text{I}^-$  (4)  $\text{CH}_3\text{COO}^-$

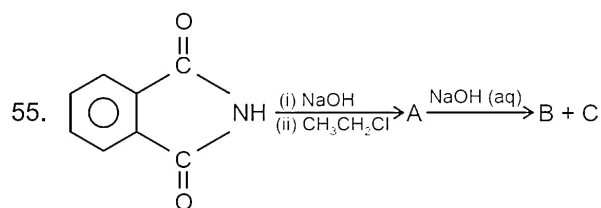
54. Consider the following reactions



Compound [A] and [B] are

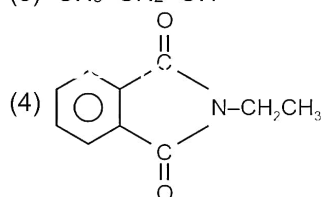
- (1)  $\text{PbI}_2$  and  $\text{K}_2[\text{PbI}_4]$  respectively  
(2)  $\text{K}_2[\text{PbI}_4]$  and  $\text{PbI}_2$  respectively  
(3)  $\text{PbI}_2$  and  $\text{K}_2[\text{PbI}_4]$  respectively  
(4)  $\text{PbI}_2$  and  $\text{K}_2[\text{PbI}_2]$  respectively

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Product B among the following can be

- (1)  $\text{CH}_3\text{CH}_2\text{C}(=\text{O})\text{NH}_2$   
 (2)  $\text{CH}_3\text{CH}_2\text{NH}_2$   
 (3)  $\text{CH}_3\text{CH}_2\text{OH}$



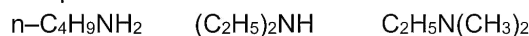
56. Consider the following statements

- (i) Insulin is globular protein  
 (ii) Globular proteins are spherical in shape  
 (iii) Myosin is present in muscles

The correct statements are

- (1) (i) and (ii) only      (2) (ii) and (iii) only  
 (3) (i) and (iii) only      (4) (i), (ii) and (iii)

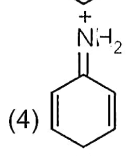
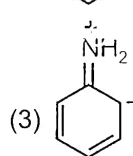
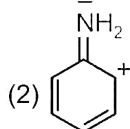
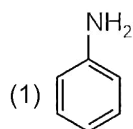
57. The correct order of boiling point for the following compounds is



(a)      (b)      (c)

- (1)  $c > b > a$       (2)  $a > b > c$   
 (3)  $b > a > c$       (4)  $c > a > b$

58. Which among the following is not a resonating structure of aniline?



59. When  $\text{K}_2\text{Cr}_2\text{O}_7$  crystals are heated with concentrated  $\text{HCl}$ , the gas evolved is

- (1)  $\text{O}_2$       (2)  $\text{Cl}_2$   
 (3)  $\text{H}_2$       (4)  $\text{HCl}$

60. The compound that does not give a blue colour in Lassaigne's test is

- (1)  $\text{NH}_2 - \text{NH}_2$       (2)  $\text{CH}_3\text{CONH}_2$   
 (3)  $\text{C}_6\text{H}_5 - \text{NH}_2$       (4)  $\text{C}_6\text{H}_5 - \text{NO}_2$

61. Identify the correct statement(s) among the following.

I. Number of chiral carbons in  $\beta\text{-D}(+)\text{glucose}$  are four.

II. Number of  $-\text{OH}$  groups present in  $\beta\text{-D}(+)\text{glucose}$  are five.

III.  $\beta\text{-D}(+)\text{glucose}$  contains pyranose ring.

- (1) II and III only      (2) II only  
 (3) I, II and III      (4) I and III only

62. The enzyme which is used to convert maltose into glucose is

- (1) Zymase      (2) Maltase  
 (3) Invertase      (4) Pepsin

63. Consider the following statements

**Statement-I:** When the polypeptide chains run parallel and are held together by hydrogen and disulphide bonds, then fibre like structure is formed.

**Statement-II:** Keratin and myosin are common examples of globular proteins.

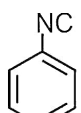
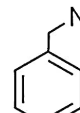
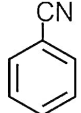
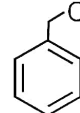
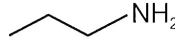
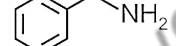
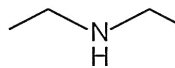
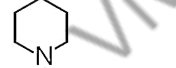
In the light of above statements, choose the correct option among the following.

- (1) Both statement-I and statement-II are true  
 (2) Both statement-I and statement-II are false  
 (3) Statement I is false but statement-II is true  
 (4) Statement I is true but statement-II is false

Space for Rough Work

64. Match the vitamin in column I with their scientific name in column II and choose the correct option.

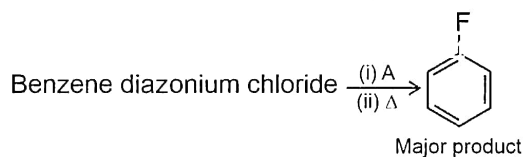
|    | Column I               |       | Column II     |
|----|------------------------|-------|---------------|
| a. | Vitamin B <sub>6</sub> | (i)   | Riboflavin    |
| b. | Vitamin C              | (ii)  | Pyridoxine    |
| c. | Vitamin B <sub>2</sub> | (iii) | Ascorbic acid |
| d. | Vitamin B <sub>1</sub> | (iv)  | Thiamine      |

- (1) a(ii), b(iii), c(i), d(iv) (2) a(ii), b(i), c(iv), d(iii)  
 (3) a(iii), b(i), c(iv), d(ii) (4) a(i), b(ii), c(iii), d(iv)
65. Which of the following bases is not present in DNA?  
 (1) Adenine (2) Uracil  
 (3) Guanine (4) Cytosine
66. When benzylamine is heated with chloroform in presence of ethanolic KOH then the major product obtained is
- (1)  (2)   
 (3)  (4) 
67. Which compound among the following does not react with Hinsberg's reagent?  
 (1)  (2)   
 (3)  (4) 
68. Consider the following statements regarding cellulose.  
 a. Cellulose is a straight chain polysaccharide composed only of  $\beta$ -D-glucose units.  
 b. In cellulose, the glycosidic linkage is between C1 of one glucose unit and C6 of the next glucose unit.

- c. Cellulose is a predominant constituent of cell wall of plant cells.

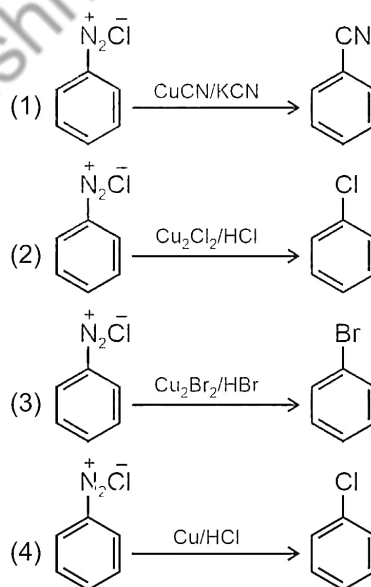
The correct statements are

- (1) a and b only (2) a and c only  
 (3) b and c only (4) a, b and c
69. Which among the following is an allylamine?  
 (1) Prop-1-en-2-amine (2) But-3-en-1-amine  
 (3) Prop-2-en-1-amine (4) But-1-en-2-amine
70. Consider the given reaction sequence



Compound A is

- (1) F<sub>2</sub> (2) CH<sub>3</sub>F  
 (3) HBF<sub>4</sub> (4) BF<sub>3</sub>
71. Which among the following reactions is not a Sandmeyer reaction?



Space for Rough Work

72. Consider the following statements.

**Statement-I:** Benzenesulphonyl chloride is used for the distinction of primary, secondary and tertiary amines.

**Statement-II:** N-Ethylbenzenesulphonamide is insoluble in alkali due to absence of acidic hydrogen atom.

In light of above statements, choose the correct option from the following.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is incorrect but statement II is correct
- (4) Statement I is correct but statement II is incorrect

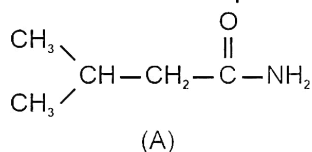
73. Non-essential amino acid among the following is

- (1) Valine
- (2) Leucine
- (3) Alanine
- (4) Lysine

74. Which of the following pair does not contain at least one fat soluble vitamin?

- (1) Vitamin A and B
- (2) Vitamin B and C
- (3) Vitamin C and D
- (4) Vitamin D and K

75. Which of the following reagent on reaction with (A) gives one carbon less in product?



- (1) Zn – Hg/HCl
- (2)  $\text{NH}_2 - \text{NH}_2/\text{OH}^-$
- (3)  $\text{LiAlH}_4/\text{H}_3\text{O}^+$
- (4)  $\text{NaOH}/\text{Br}_2$

76.  $\alpha$  and  $\beta$  forms of glucose are called

- (1) Enantiomers
- (2) Anomers
- (3) Conformers
- (4) Metamers

77. The violet coloured compound formed when sulphur containing sodium fusion extract is treated with sodium nitroprusside is

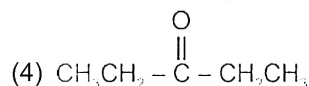
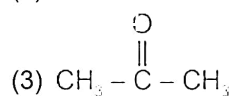
- (1)  $[\text{Fe}(\text{CN})_5\text{NOS}]^{4-}$
- (2)  $[\text{Fe}(\text{CN})_5\text{NO}]^{2-}$
- (3)  $[\text{Fe}(\text{CN})_3\text{NOS}]^{2-}$
- (4)  $[\text{Fe}(\text{CN})_5\text{SCN}]^{4-}$

78. The number of moles of  $\text{KMnO}_4$  required to completely oxidise one mole of  $\text{FeSO}_4$  in acidic medium is

- (1) 5
- (2)  $\frac{1}{5}$
- (3)  $\frac{2}{5}$
- (4) 1

79. The compound which will not form iodoform on reaction with  $\text{I}_2/\text{NaOH}$  is

- (1)  $\text{CH}_3\text{CHO}$
- (2)  $\text{CH}_3\text{CH}_2\text{OH}$



- (4)  $\text{CH}_3\text{CH}_2 - \text{C} - \text{CH}_2\text{CH}_3$

80. The working pH range of phenolphthalein indicator is

- (1) 3.1 to 4.5
- (2) 6.2 to 8.2
- (3) 4.2 to 6.2
- (4) 8.2 to 10.2

81. Match the column-I with column-II and choose the correct option.

|    | Column-I  |       | Column-II        |
|----|-----------|-------|------------------|
| a. | Group-VI  | (i)   | $\text{Fe}^{3+}$ |
| b. | Group-V   | (ii)  | $\text{Mg}^{2+}$ |
| c. | Group-I   | (iii) | $\text{Sr}^{2+}$ |
| d. | Group-III | (iv)  | $\text{Pb}^{2+}$ |

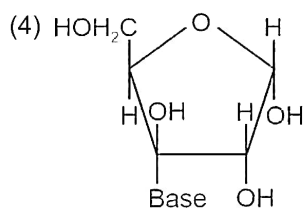
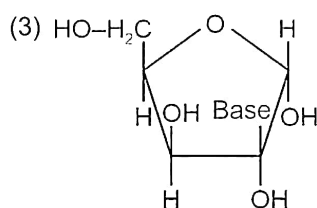
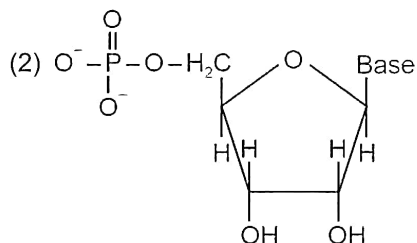
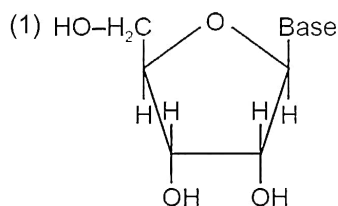
- (1) a(ii), b(iii), c(iv), d(i)
- (2) a(i), b(iii), c(iv), d(ii)
- (3) a(iv), b(ii), c(iii), d(i)
- (4) a(ii), b(i), c(iv), d(iii)

82. Positively charged sol among the following is

- (1) Silver sol
- (2)  $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
- (3) Congo red sol
- (4) Gelatin

Space for Rough Work

83. The correct structure of nucleoside is



84. Which of the following sulphides is yellow in colour?

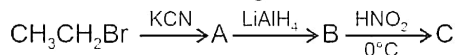
- (1) CdS (2) CoS  
(3) ZnS (4) NiS

85. The indicator used in iodometric titrations is

- (1) Starch  
(2) Phenolphthalein  
(3) Methyl orange  
(4) Phenol red

### SECTION-B

86. Consider the following reaction



The correct structure of A, B and C respectively are

- (1)  $\text{CH}_3\text{CH}_2\text{NC}$ ,  $\text{CH}_3\text{CH}_2\text{NHCH}_3$  and  $\text{CH}_3\text{CH}_2\text{NO}_2$   
(2)  $\text{CH}_3\text{CH}_2\text{CN}$ ,  $\text{CH}_3\text{CH}_2\text{NH}_2$  and  $\text{CH}_3\text{CH}_2\text{OH}$   
(3)  $\text{CH}_3\text{CH}_2\text{NC}$ ,  $\text{CH}_3\text{CH}_2\text{NH}_2$  and  $\text{CH}_3\text{CH}_2\text{NHOH}$   
(4)  $\text{CH}_3\text{CH}_2\text{CN}$ ,  $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$  and  $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$

87. Given below are two statements one is labelled as assertion (A) and other is labelled as reason (R)

**Assertion (A):** Aniline does not undergo Friedel-Crafts reaction.

**Reason (R):** Aniline reacts with aluminium chloride and nitrogen of aniline acquires positive charge and hence acts as strong deactivating group.

In the light of above statements, choose the most appropriate answer from the options given below.

- (1) Both (A) and (R) are true and the (R) is the correct explanation of (A)  
(2) Both (A) and (R) are true but the (R) is not the correct explanation of (A)  
(3) (A) is true but (R) is false  
(4) Both (A) and (R) are false

88. Phosphodiester linkage is present in

- (1) Nucleic acids (2) Proteins  
(3) Vitamins (4) Carbohydrates

89. Cheilosis is caused by deficiency of

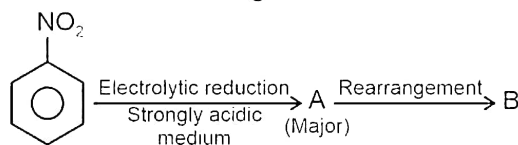
- (1) Vitamin B<sub>1</sub> (2) Vitamin B<sub>2</sub>  
(3) Vitamin B<sub>6</sub> (4) Vitamin B<sub>12</sub>

90. Starch is a polymer of

- (1)  $\beta$ -D glucose (2)  $\alpha$ -D glucose  
(3)  $\beta$ -D Galactose (4)  $\beta$ -D fructose

Space for Rough Work

91. Consider the following reaction



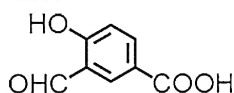
Product B is

- (1)
- (2)
- (3)
- (4)

92.  $\text{Ag}^+$ ,  $\text{Hg}_2^{2+}$  and  $\text{Pb}^{2+}$  are grouped together in first group because they form insoluble

- (1) Sulphates (2) Chlorides  
(3) Hydroxides (4) Carbonates

93. Consider the statements given below about the given compound.



- a. It gives positive test with  $\text{NaHCO}_3$ .  
b. It gives positive test with 2,4-DNP.  
c. It gives positive test with Hinsberg's reagent.

The correct statement(s) is/are

- (1) a and b only (2) b and c only  
(3) b only (4) a, b and c

94. Given below are the two statements one is labelled as **Assertion (A)** and other is labelled as **Reason (R)**.

**Assertion (A):** Aromatic amines are weaker bases than ammonia.

**Reason (R):** In aromatic amines, the aryl group behaves as an electron withdrawing group.

In light of the above statements, choose the correct option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)  
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)  
(3) (A) is false but (R) is true  
(4) (A) is true but (R) is false

95. When  $\text{H}_2\text{S}$  passed through a mixture containing  $\text{Cu}^{2+}$ ,  $\text{Ni}^{2+}$ ,  $\text{Zn}^{2+}$  in dilute acidic solution then which of the following ion will precipitate?

- (1)  $\text{Cu}^{2+}$ ,  $\text{Ni}^{2+}$  only (2)  $\text{Ni}^{2+}$  only  
(3)  $\text{Cu}^{2+}$ ,  $\text{Zn}^{2+}$  only (4)  $\text{Cu}^{2+}$  only

96. Consider the following statements

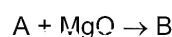
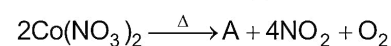
**Statement-I:** During the formation of p-nitro acetanilide from acetanilide, electrophilic substitution reaction takes place.

**Statement-II:** Acetanilide group acts as an electron donating group to benzene ring.

Choose the correct option.

- (1) Both statement I and statement II are correct  
(2) Both statement I and statement II are incorrect  
(3) Statement I is incorrect but statement II is correct  
(4) Statement I is correct but statement II is incorrect

97. Consider the following reactions



The colour of compound B is

- (1) Green (2) Blue  
(3) Pink (4) White

Space for Rough Work

98. Which among the following match is incorrect regarding colour of metal ions?

|     | Cations          |   | Colour       |
|-----|------------------|---|--------------|
| (1) | $\text{Cu}^{2+}$ | – | Blue         |
| (2) | $\text{Co}^{2+}$ | – | Yellow       |
| (3) | $\text{Ni}^{2+}$ | – | Bright green |
| (4) | $\text{Mn}^{2+}$ | – | Light pink   |

99. Given below are the two statements one is labelled as **Assertion (A)** and other is labelled as **Reason (R)**.

**Assertion (A):** A white gelatinous precipitate of aluminium hydroxide is soluble in excess of sodium hydroxide.

**Reason (R):** Formation of sodium aluminate takes place on reaction of  $\text{Al}(\text{OH})_3$  with excess of  $\text{NaOH}$ .

In light of the above statements, choose the correct option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)  
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)  
 (3) (A) is false but (R) is true  
 (4) (A) is true but (R) is false
100. Which among the following cations gives apple green colour to the flame test?

- (1)  $\text{Cu}^{2+}$  (2)  $\text{Sr}^{2+}$   
 (3)  $\text{Ca}^{2+}$  (4)  $\text{Ba}^{2+}$

## BOTANY

### SECTION - A

101. Which among the following is considered as broadly utilitarian services that nature provides?

- (1) Industrial product (2) Pollination  
 (3) Foods (4) Fibre

102. Recent extinction of Thylacine was from the region of

- (1) Africa (2) Mauritius  
 (3) Australia (4) Russia

103. \_\_\_\_\_% of photosynthetically active radiation is being captured by plants in synthesis of organic matter.

Choose the option to **correctly** fill in the blank.

- (1) >50 (2) 12-20  
 (3) <0.1 (4) 2-10

104. Earthworms are

- (1) Detritivores (2) Parasites  
 (3) Producers (4) Top carnivores

105. Secondary succession

- (1) Starts at the area where no living organism ever existed.  
 (2) Takes longer time than primary succession to reach climax  
 (3) Shows that establishment of pioneer communities is more easier.  
 (4) Starts at the barren area

106. Read the following statements and choose the **correct** option.

**Statement A:** Standing state is a certain mass of living material present at each trophic level at a particular time.

**Statement B:** According to Robert Costanza and his colleagues, the cost of climate regulation and habitat for wildlife are about 6 percent each of the total cost of ecosystem services.

- (1) Both statements A and B are incorrect  
 (2) Both statements A and B are correct  
 (3) Only statement A is correct  
 (4) Only statement B is correct

Space for Rough Work

107. Match List-I with List-II.

| List-I                          | List-II                      |
|---------------------------------|------------------------------|
| (a) Sacred Groves               | (i) Rio de Janeiro, 1992     |
| (b) <i>Ex situ</i> conservation | (ii) Wildlife safari parks   |
| (c) Earth summit                | (iii) Aravalli Hills         |
| (d) Hotspots                    | (iv) High degree of Endemism |

Choose the **correct** answer from the options given below:

- (1) (a) (iii), (b) (iv), (c) (i), (d) (ii)
- (2) (a) (iii), (b) (ii), (c) (i), (d) (iv)
- (3) (a) (ii), (b) (iv), (c) (iii), (d) (i)
- (4) (a) (iv), (b) (iii), (c) (ii), (d) (i)

108. Arrange the following major taxa of invertebrates in ascending order w.r.t. number of species present globally and choose the **correct** option.

- (a) Molluscs
- (b) Insects
- (c) Crustaceans
- (1) (b) < (c) < (a)
- (2) (a) < (b) < (c)
- (3) (c) < (b) < (a)
- (4) (c) < (a) < (b)

109. Tropical regions are rich in biodiversity than temperate regions. One of the reasons is that, the former

- (1) Areas have undergone frequent glaciation in the past
- (2) Have less humid and cold temperature
- (3) Have constant and predictable environment
- (4) Show least niche specialisation

110. Read the following statements and select the **correct** option.

**Assertion (A):** Humus undergoes decomposition at an extremely slow rate.

**Reason (R):** There is no need of oxygen in the process of decomposition that forms humus as it is controlled mainly by chemical composition of detritus.

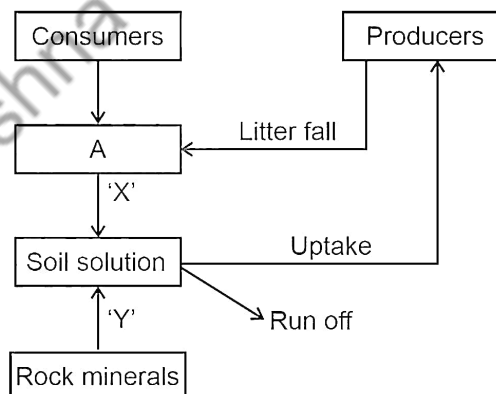
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true but (R) is false.
- (4) Both (A) and (R) are false.

111. In Species-Area relationships equation  $\log S = \log C + Z \log A$

S represents

- (1) Area in square km
- (2) Species richness
- (3) Regression coefficient
- (4) Y intercept

112.



Identify the 'X', 'Y' and 'A' in above flow-chart and mark the option with their **incorrect** match.

- (1) The substance represented by A is broken down by earthworms.
- (2) Process represented by X is performed by microbes
- (3) Process Y is performed by detritivores
- (4) A represents detritus

Space for Rough Work

113. In an aquatic ecosystem all of the following are producers, **except**
- Phytoplanktons
  - Algae
  - Floating plants
  - Hydra* and Corals
114. National aquatic animal of India is
- Gangetic shark
  - River dolphin
  - Sea horse
  - Blue whale
115. About 71% of the total global carbon is
- Found in atmosphere
  - Found dissolved in oceans
  - Present in microbes
  - Found in dead organisms
116. Consider the following food chain.  
Grass → Grasshopper → Frog → Snake  
If 100 kJ energy is converted into biomass by producers, then energy available to the frog in the form of food will be
- 1000 J
  - 100 J
  - 10 J
  - 1 J
117. Which of the given is **not** an exotic species in India?
- Parthenium*
  - Lantana*
  - Eichhornia*
  - Nile perch
118. What percentage of earth's land area is covered by the 34 hotspots?
- Only 8%
  - Less than 2%
  - About 50%
  - More than 10%
119. Who popularized the term to describe the combined diversity at the levels of biological organization?
- Edward Wilson
  - Alexander von Humboldt
  - David Tillmann
  - Paul Ehrlich
120. Select the **correct** relationship between net primary productivity (NPP) and gross primary productivity.
- $NPP = \text{Gross primary productivity} \times \text{Respiratory losses}$
  - $NPP = \text{Gross primary productivity} - \text{Respiratory losses}$
  - $NPP = \text{Gross primary productivity} + \text{Respiratory losses}$
  - $NPP = \frac{\text{Gross Primary Productivity}}{\text{Respiratory losses}}$
121. If a tree supports large number of herbivorous birds which are eaten by hawks of that area, what shape of pyramid of number is obtained in this condition?
- Upright
  - Spindle shaped
  - Both upright or inverted
  - Inverted only
122. Pioneer community in pond ecosystem is formed by
- Lichens
  - Shrubs
  - Phytoplanktons
  - Bryophytes
123. For very large areas, such as entire continent, the value of regression coefficient obtained from species-area relationship curve is
- 1.3 to 1.5
  - 0.1 to 0.2
  - 0.3 to 0.5
  - 0.6 to 1.2
124. India has 'A' biosphere reserves and 'B' National Parks.  
Choose the **correct** option for A and B.
- A-14, B-90
  - A-25, B-34
  - A-448, B-34
  - A-90, B-448

Space for Rough Work

125. Select the **correct** statement.

- (1) *Rauwolfia vomitoria* results in the variations in potency and concentration of active chemical due to species diversity.
- (2) The effect of reduction in biodiversity has been explained by Paul Ehrlich through Rivet popper hypothesis.
- (3) Passenger pigeon has become extinct due to alien species invasion.
- (4) Speciation is generally not a function of time.

126. Which among the following characteristic is **not** true for the organisms susceptible to extinction?

- (1) Large body size
- (2) Small population size
- (3) Lower trophic level in food chain
- (4) Low rate of reproduction

127. Which among the following statements is **false** w.r.t. primary productivity?

- (1) It refers to the rate at which sunlight is captured by producers for synthesis of food during photosynthesis.
- (2) It refers to the rate at which biomass or organic matter is produced by plants.
- (3) It is available biomass for the consumption to carnivores.
- (4) It can be further divided into two categories, i.e., gross primary productivity and net primary productivity.

128. Carbon dioxide can be added into the atmosphere from all of the following processes, **except**

- (1) Burning of forests and organic debris
- (2) Combustion of fossil fuel in vehicles
- (3) Respiration and decomposition
- (4) Photosynthesis in higher plants

129. State **True (T)** or **False (F)** for the given statements.

- (A) Saprophytes and detritivores are not given any place in ecological pyramids.
- (B) Annual net primary productivity of whole biosphere is approximately 170 million tons.
- (C) Energy at a higher trophic level is always more than at a lower trophic level.
- (D) The function of reservoir is to meet with deficit which occurs due to imbalance in rate of influx and efflux.

Choose the **correct** option.

(A) (B) (C) (D)

- (1) F T F F
- (2) F F F T
- (3) T F F T
- (4) T T T F

130. Choose the option w.r.t number of estimated species found in the tropical Amazonian rain forest in South America.

|     | Birds              | Reptiles | Fish             | Amphibians |
|-----|--------------------|----------|------------------|------------|
| (1) | 378                | 3,000    | 1,300            | 1300       |
| (2) | 1,300              | 378      | 3,000            | 427        |
| (3) | 427                | 300      | More than 40,000 | 300        |
| (4) | More than 1,25,000 | 378      | 1,300            | 427        |

131. India has more than (i) species of birds because most of the land area of our country lies in (ii). Select the **correct** option to fill in the blanks (i) and (ii).

(i) (ii)

- (1) 40,000 Tropics
- (2) 35,000 Temperate areas
- (3) 18,000 Temperate areas
- (4) 1,200 Tropics

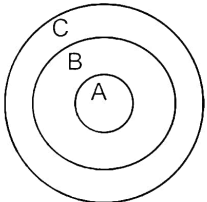
Space for Rough Work

132. The third stage of transitional communities in hydrarch succession is
- (1) Scrub stage
  - (2) Submerged plant stage
  - (3) Reed-swamp stage
  - (4) Marsh-meadow stage
133. Increase in biodiversity of an ecosystem contribute to
- (1) Increase in productivity
  - (2) Decrease in stability of the ecosystem
  - (3) High variation in total biomass from year to year
  - (4) Increase in susceptibility towards alien species invasions
134. Mark the statements as **true (T)** or **false (F)** and select the **correct** option.
- a. Strict protection of biodiversity hotspots could reduce the ongoing mass extinctions by almost 30%.
  - b. Amazonian rain forest contributes 20% of total oxygen in the earth's atmosphere.
- (1) a(T); b(F)                      (2) a(T); b(T)
  - (3) a(F); b(T)                      (4) a(F); b(F)
135. Read the statements A and B and select the **correct** option for them.
- Statement A:** Globally among the animal groups, the most species rich taxonomic group is of vertebrates.
- Statement B:** Globally among plants, the diversity of mosses is more than angiosperms.
- (1) Only statement A is correct
  - (2) Only statement B is correct
  - (3) Both A and B statements are correct
  - (4) Both A and B statements are incorrect

**SECTION - B**

136. Red data book contains information on
- (1) Threatened species
  - (2) Aquatic vertebrates only
  - (3) Economically important plants
  - (4) Only extinct species
137. In a terrestrial food chain, cow is
- (1) Tertiary consumer
  - (2) Secondary consumer
  - (3) Primary consumer
  - (4) Primary producer
138. Read the following statements and choose the **correct** option.
- Statement (A):** The Western ghats have a greater amphibian species diversity than the Eastern Ghats.
- Statement (B):** Bioprospecting is exploring molecular, genetic and species-level diversity for products of economic importance.
- (1) Both statements A and B are correct
  - (2) Only statement B is correct
  - (3) Only statement A is correct
  - (4) Both statements A and B are incorrect
139. Which of the following are ecosystem services?
- (a) Generating fertile soils
  - (b) Providing wild life habitat
  - (c) Mitigating droughts and floods
  - (d) Providing storage site for carbon
- The **correct** ones are
- (1) (a) and (b) only                      (2) (c) and (d) only
  - (3) (b) and (d) only                      (4) All (a), (b), (c) & (d)

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140. Read the following statements and select the **correct** option.  
**Assertion (A):** Pyramid of energy can be inverted in most of the ecosystems.  
**Reason (R):** The flow of energy is bidirectional from producer to consumer level, and vice versa.  
 (1) Both (A) and (R) are true and (R) is the correct explanation of (A)  
 (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)  
 (3) Both (A) and (R) are false  
 (4) (A) is true but (R) is false.
141. India has only 2.4% of world's land area and its share to the global species diversity is  
 (1) 8.1% (2) 21%  
 (3) 22% (4) 62%
142. The phenomenon of incorporation of nutrients in living microbes is called  
 (1) Mineralisation  
 (2) Decomposition  
 (3) Nutrient immobilisation  
 (4) Productivity
143. How many of the following changes occur(s) during the process of ecological succession?  
 (a) Total biomass increases  
 (b) Decrease in diversity  
 (c) Dry condition to mesic condition  
 (d) Decrease in humus content  
 (1) Three (2) Two  
 (3) Four (4) One
144. Select the **incorrect** one w.r.t. characteristics of anthropogenic ecosystems.  
 (1) Possess self regulatory mechanism  
 (2) Have little diversity  
 (3) Simple food chain  
 (4) High productivity
145. Which of the following relations can be derived from Tilman's experiment?  
 (1) Species richness  $\propto$  Explored area  
 (2) Productivity  $\propto$  Extent of diversity  
 (3) Habitat loss  $\propto$  Loss of diversity  
 (4) Over-exploitation  $\propto$  Co-extinction
146. Primary production in the ecosystem is expressed as  
 (1)  $\text{g m}^{-2}$  (2)  $\text{kcal g}^{-2}$   
 (3)  $\text{g m}^2 \text{yr}^{-1}$  (4)  $\text{kcal m}^{-3}$
147. Which among the following is **not** a part of 'The Evil Quartet' regarding biodiversity losses?  
 (1) Habitat loss and fragmentation  
 (2) Over-exploitation  
 (3) Alien species invasion  
 (4) Co-evolution
148. Different zones in a terrestrial biosphere reserve are labelled as **A, B** and **C** as shown below.
- 
- In which zone(s), human settlements of local people can be seen?  
 (1) B only (2) B and C only  
 (3) C only (4) A only
149. Greatest biodiversity on earth is found in  
 (1) Amazonian rainforests  
 (2) Temperate forest of India  
 (3) Western ghats of India  
 (4) Polar regions
150. Third trophic level of a food chain can be occupied by all of the following, **except**  
 (1) Secondary consumers  
 (2) Carnivores  
 (3) Animal feeding on plant eating organisms  
 (4) Herbivores

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**ZOOLOGY**

## SECTION - A

151. Which of the following enzymes is responsible for restricting the growth of bacteriophages in *E. coli*?
- (1) DNA ligase
  - (2) Lysozyme
  - (3) Restriction endonuclease
  - (4) DNA polymerase
152. Significance of 'heat shock' method in bacterial transformation is to facilitate
- (1) Binding of DNA to peptidoglycan present in the bacterial cell wall
  - (2) Uptake of DNA through transient pores present in the bacterial cell wall
  - (3) Uptake of DNA through transport proteins present in the bacterial cell wall
  - (4) The decrease in size of pores present in the bacterial cell wall
153. Choose the **correct** option to complete the analogy w.r.t. pBR322.
- Bam*H I : tet<sup>R</sup> :: \_\_\_\_ : amp<sup>R</sup>
- (1) *Pvu* II
  - (2) *Cla* I
  - (3) *Sal* I
  - (4) *Pvu* I
154. **Assertion (A):** Bacteria are immune to their own restriction enzymes, even if they contain the specific recognition sequences.
- Reason (R):** Bacteria have an enzyme which adds methyl groups to their own DNA.
- In the light of above statements, select the **correct** option.
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
  - (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
  - (3) (A) is true; (R) is false
  - (4) Both (A) and (R) are false

155. Transgenic models exist for how many human diseases given in the box below?

Alzheimer's disease, Cystic fibrosis, Cancer, Rheumatoid arthritis

Select the **correct** option.

- (1) One                      (2) Four  
(3) Two                     (4) Three

156. Select the **incorrect** match.

|     |              |   |            |
|-----|--------------|---|------------|
| (1) | Lepidopteran | – | Armyworms  |
| (2) | Coleopteran  | – | Beetles    |
| (3) | Dipteran     | – | Flies      |
| (4) | Lepidopteran | – | Mosquitoes |

157. Animals that have had their DNA manipulated to possess and express a foreign gene can be used for

- a. Study of diseases
- b. Testing vaccine safety
- c. Investigation of a new treatment for autoimmune diseases
- d. Obtaining useful biological products

Select the **correct** option.

- (1) a and b only                      (2) b and d only  
(3) a, b, c and d                    (4) d only

158. Plasmid DNA is **not**

- (1) Capable of independent replication
- (2) Circular and double stranded
- (3) A chromosomal DNA
- (4) Use as a vector

159. The process through which a piece of DNA is introduced in a host bacterium is known as

- (1) PCR
- (2) Transformation
- (3) Competency
- (4) Gel electrophoresis

### Space for Rough Work

160. The night blindness caused due to vitamin-A deficiency can be treated by taking nutritionally enhanced food. This can be done by consumption of
- Bt potato
  - Transgenic tobacco plant
  - Golden rice
  - Bt tomato
161. Which of the following is a desirable feature of an ideal cloning vector?
- Absence of *ori* sequence
  - Presence of many recognition sites for a single restriction enzyme
  - Large in size
  - Presence of a selectable marker
162. The decisions regarding the validity of GM research and the safety of introducing GM-organisms for public services in India are made by
- CDRI
  - EFB
  - GEAC
  - ICMR
163. Which of the following given sequences is acted upon by *EcoRI*?
- $$\begin{array}{c} \downarrow \\ 5' - \text{GAATTC} - 3' \\ 3' - \text{CTTAAG} - 5' \\ \uparrow \end{array}$$
  - $$\begin{array}{c} \downarrow \\ 5' - \text{GGATCC} - 3' \\ 3' - \text{CCTAGG} - 5' \\ \uparrow \end{array}$$
  - $$\begin{array}{c} \downarrow \\ 5' - \text{CCCGGG} - 3' \\ 3' - \text{GGGCCC} - 5' \\ \uparrow \end{array}$$
  - $$\begin{array}{c} \downarrow \\ 5' - \text{AAGCTT} - 3' \\ 3' - \text{TTCGAA} - 5' \\ \uparrow \end{array}$$
164. How many DNA fragments are produced from a B-DNA after 5 cycles of PCR?
- Sixteen
  - Thirty two
  - Eight
  - Sixty four
165. Select the incorrect statement among the following.
- The separated DNA fragments after gel electrophoresis can be visualised after staining the DNA with ethidium bromide followed by exposure to UV radiation.
  - Elution is the process by which separated bands of DNA are cut out from the agarose gel and extracted from the gel piece.
  - Bacteriophages as compared to plasmids have high copy number of their genome within the bacterial cells.
  - In PCR, extension of primers occurs from the 5' OH-end of growing strand.
166. Read the following statements.
- Statement (A):** Sticky ends facilitate the formation of hydrogen bonds with their complementary cut counterparts.
- Statement (B):** Restriction enzymes cut the phosphodiester bonds present between the two nucleotides in a polynucleotide chain.
- Select the **correct** option.
- Both statements (A) and (B) are correct
  - Both statements (A) and (B) are incorrect
  - Only statement (A) is correct
  - Only statement (B) is correct
167. A simple stirred-tank bioreactor comprises all of the following components, **except**
- An agitator system
  - Flat bladed impellers
  - A sparger
  - A foam breaker
168. Genetically engineered insulin is useful over insulin isolated from non-human sources as
- Latter does not induce unwanted immunological responses
  - Large scale production of former is not possible
  - Former increases the risk of transmission of animal diseases to humans
  - Former does not induce unwanted immunological responses

Space for Rough Work

169. DNA threads which have been precipitated by addition of chilled ethanol during their isolation are removed by
- RNA interference
  - Splicing
  - Spooling
  - Elution
170. In a continuous culture system
- Fresh medium is not added to the culture
  - Cells remain in physiologically inactive state throughout the process
  - Larger biomass is produced as compared to closed culture system
  - Used medium is recycled to obtain maximum yield
171. All of the following are methods of direct gene transfer, **except**
- Micro-injection
  - Biolistics
  - Electroporation
  - Disarmed retrovirus mediated
172. Which of the following statements does not belong to the three critical research areas of biotechnology?
- Providing the best catalyst in the form of improved organism usually a microbe or pure enzyme.
  - Creating optimal conditions through engineering for a catalyst to act.
  - To file a biopatent on the developed novel research.
  - Downstream processing technologies to purify the protein/organic compound.
173. All of the following are required for performing PCR, **except**
- Template DNA
  - dUTPs
  - Taq* polymerase
  - Primers
174. The host used by Eli Lilly to produce the recombinant insulin was
- Agrobacterium tumefaciens*
  - Bacillus thuringiensis*
  - Meloidogyne incognita*
  - Escherichia coli*
175. Consider the following :
- 'A' peptide
  - 'B' peptide
  - 'C' peptide
- How many of the above mentioned chain(s) is/are absent in proinsulin?
- One
  - Two
  - Three
  - Zero
176. The proteins encoded by the gene X controls corn borer and that of Y and Z control the cotton bollworms.
- Select the **correct** option to fill in the blanks.
- |     | X              | Y              | Z              |
|-----|----------------|----------------|----------------|
| (1) | <i>cryIIAb</i> | <i>cryIAb</i>  | <i>cryIAc</i>  |
| (2) | <i>cryIAc</i>  | <i>cryIIAb</i> | <i>cryIAb</i>  |
| (3) | <i>cryIAb</i>  | <i>cryIAc</i>  | <i>cryIIAb</i> |
| (4) | <i>cryIAc</i>  | <i>cryIIAb</i> | <i>cryIIAb</i> |
177. RNAi involves formation of which of the following molecules that binds to and prevents the translation of the target mRNA?
- dsDNA
  - dsRNA
  - ssRNA
  - ssDNA
178. **Assertion (A):** Insulin is not taken orally by the diabetics.
- Reason (R):** Insulin is a peptide hormone that gets cleaved off by proteases present in the digestive juices before reaching the blood stream.
- In the light of above statements, select the **correct** option.
- Both (A) and (R) are true but (R) is not the correct explanation of (A)
  - (A) is true; (R) is false
  - Both (A) and (R) are false
  - Both (A) and (R) are true and (R) is the correct explanation of (A)

Space for Rough Work

179. How many intra-chain disulphide bond(s) is/are present in pro-insulin and insulin respectively?

- (1) One, Two (2) Two, One  
(3) Three, Three (4) One, One

180. Select the **incorrect** match among the following.

|     |                |   |                                                                 |
|-----|----------------|---|-----------------------------------------------------------------|
| (1) | Serum analysis | – | Conventional method of disease diagnosis                        |
| (2) | ELISA          | – | Based on antigen-antibody interaction                           |
| (3) | PCR            | – | <i>In vivo</i> method of gene cloning                           |
| (4) | Probe          | – | A single stranded DNA or RNA tagged with a radioactive molecule |

181. In gene therapy for ADA deficiency, functional ADA cDNA is introduced in 'X'. Identify 'X' and select the **correct** option w.r.t it.

- (1) An agranulocyte which constitutes 20-25% of total WBCs in humans.  
(2) A granulocyte which constitutes 0.5-1% of total WBCs in humans.  
(3) An agranulocyte which constitutes 6-8% of total WBCs in humans.  
(4) A granulocyte and the most abundant WBC in humans.

182. The toxin released by *B. thuringiensis* is able to kill certain insects but does not kill the bacteria itself because

- (1) Bt toxin does not get activated as the gut of bacteria has alkaline pH.  
(2) Bt toxin protein exists as an inactive protoxin in the bacteria.  
(3) Bt toxin is not fatal to bacteria but makes them weak.  
(4) Bt toxin is formed only after the death of the bacteria.

183. A probe is designed to differentiate between cells having mutated gene from cells having non-mutated gene.

In a cluster of cells, detection by autoradiography revealed that clones having mutated gene do not appear on the photographic film. This happened because

- (1) Probe binds to non-mutated gene but did not appear.  
(2) Probe has a complementarity with the mutated gene.  
(3) Probe has no complementarity with the mutated gene.  
(4) Probe has no complementarity with the non-mutated gene.

184. Match column I with column II.

|     | Column I                               |       | Column II                                  |
|-----|----------------------------------------|-------|--------------------------------------------|
| (a) | $\alpha$ -1-antitrypsin                | (i)   | Prepared by an American company, Eli Lilly |
| (b) | Human alpha lactalbumin                | (ii)  | Used to treat emphysema                    |
| (c) | Recombinant insulin                    | (iii) | Basmati rice                               |
| (d) | 27 documented varieties grown in India | (iv)  | Present in milk produced by 'Rosie'        |

Select the **correct** option.

- (a) (b) (c) (d)  
(1) (i) (ii) (iii) (iv)  
(2) (ii) (iii) (iv) (i)  
(3) (iii) (i) (ii) (iv)  
(4) (ii) (iv) (i) (iii)

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185. The term which refers to the use of bio-resources by organisations with no proper authorisation from the people concerned and without compensatory payment is called

- (1) Bioethics
- (2) Biopiracy
- (3) Biosafety
- (4) Biodegradation

### SECTION - B

186. A student wants to isolate the genetic material of a fungal cell. Which of the following combination of enzymes must be used by him?

- (1) Lysozyme, DNase, Ribonuclease
- (2) Chitinase, Ribonuclease, Protease
- (3) Chitinase, DNase, Protease
- (4) DNase, Lysozyme, Chitinase

187. Arrange the following steps in the correct sequence while genetically modifying an organism.

- (a) Modification in a vector with same restriction enzyme as used for desired DNA.
- (b) Maintenance of introduced DNA in the host cell.
- (c) Identification and isolation of desired DNA.
- (d) Introduction of desired DNA into the host cell.
- (e) Transfer of desired DNA to progeny of host.

Select the **correct** option.

- (1) (c) → (a) → (e) → (d) → (b)
- (2) (a) → (c) → (e) → (b) → (d)
- (3) (c) → (a) → (d) → (b) → (e)
- (4) (a) → (c) → (d) → (e) → (b)

188. Which of the following sets of restriction enzymes produce sticky ends?

- (1) *Bam*H I and *Eco* RV
- (2) *Eco*R I and *Bam*H I
- (3) *Hind* II and *Eco*R I
- (4) *Eco* RV and *Pst* I

189. Consider the following statements.

- (A) It is a pathogen of several dicot plants.
- (B) It delivers T-DNA into plants and convert normal cells into tumor cells.

Select the organism for which the above-mentioned features are true.

- (1) *Salmonella typhimurium*
- (2) *Agrobacterium tumefaciens*
- (3) Retrovirus
- (4) *Meloidogyne incognita*

190. The enzymes responsible for restricting the growth of bacteriophages in *E. coli* belong to the same class as that of the enzyme characterised by the enzyme commission number

- (1) 2.4.3.1
- (2) 4.3.2.6
- (3) 3.6.1.2
- (4) 5.7.3.2

191. Quality control testing of a product prepared from biotechnological principles is placed under

- (1) Upstream processing
- (2) Biosynthetic stage
- (3) Downstream processing
- (4) Bioprocess engineering

192. All of the following are true for bioreactors, **except**

- (1) It is a vessel in which raw materials are biologically converted into specific products.
- (2) It provides optimal conditions for achieving the desired products.
- (3) Large volumes of approximately 100-1000 L of culture can be processed in it.
- (4) Oxygen bubbles dramatically increase the oxygen transfer area in a simple stirred-tank bioreactor.

Space for Rough Work

193. The plant part used for tissue culture is sterilized by
- (1) Anti-microbial chemicals
  - (2) Ionising radiations
  - (3) Nutrient media
  - (4) Non-ionising radiations
194. The first human hormone produced by the recombinant DNA technology is
- (1) Insulin
  - (2) Melatonin
  - (3) Testosterone
  - (4) Progesterone
195. A pathogenic DNA molecule in small amount was extracted from a blood sample. Which of the following techniques should be used to amplify the DNA found in the given sample?
- (1) PCR
  - (2) Western blotting
  - (3) Northern blotting
  - (4) ELISA
196. If 'ori' from a plasmid of an organism is replaced with an alien piece of DNA, then the consequence of this experiment will be that
- (1) The alien piece of DNA will replicate.
  - (2) The alien piece of DNA will not be able to replicate.
  - (3) The alien piece of DNA will be produced with high copy numbers.
  - (4) The chromosomal DNA of the organism will provide genes for replication of the ligated alien DNA.
197. In a gene cloning experiment, a foreign gene encoding enzyme  $\beta$ -galactosidase is inserted at the site same as that of  $amp^R$  in vector pBR322, followed by its insertion in the host cell.
- Select the **incorrect** statement w.r.t the obtained results after the completion of the above-mentioned experiment.
- (1) Recombinants give blue coloured colonies in the presence of a chromogenic substrate and will not grow in ampicillin containing medium.
  - (2) Non-recombinants will give white coloured colonies in the presence of a chromogenic substrate.
  - (3) Non-recombinant transformants will be ampicillin and tetracycline resistant.
  - (4) Non-transformants will be ampicillin sensitive but resistant to tetracycline.
198. Read the following statements and select the **correct** option.
- Statement (A):** DNA molecules are lipophilic in nature.
- Statement (B):** In gel electrophoresis, the larger the DNA fragment the farther it moves from the cathode.
- (1) Only statement (A) is correct.
  - (2) Both statements (A) and (B) are incorrect.
  - (3) Only statement (B) is correct.
  - (4) Both statements (A) and (B) are correct.

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199. **Assertion (A):** *Hind* II is the first restriction endonuclease to be isolated, not *Hind* I.

**Reason (R):** Each restriction endonuclease recognises a specific palindromic nucleotide sequence in the DNA.

In the light of above statements, select the **correct** option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true; (R) is false
- (4) Both (A) and (R) are false

200. The DNA 'X' and 'Y' are treated with the same restriction enzyme. Both DNA have the same number of restriction sites for the used restriction enzyme. After performing the restriction digestion experiment, 3 fragments of 'X' DNA and 2 fragments of 'Y' DNA were observed. Select the **correct** option w.r.t. the shape of 'X' and 'Y'.

'X'

'Y'

- |              |          |
|--------------|----------|
| (1) Circular | Linear   |
| (2) Linear   | Circular |
| (3) Linear   | Linear   |
| (4) Circular | Circular |

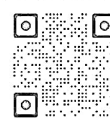
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